Design, Implementation, and Privacy-Preserving Deployment of an Electroencephalography (EEG) Motor Imagery Brain-Computer Interface (BCI)

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Abstract

This report presents the end-to-end design, implementation, evaluation, and proposed enterprise deployment architecture of a non-invasive Motor Imagery Brain-Computer Interface (BCI). Utilizing multi-run Electroencephalography (EEG) data from the PhysioNet EEGMMIDB database ($N = 90$ trials per subject), the system classifies imagined left-hand versus right-hand movements. Critical machine learning bottlenecks—including sample starvation, high-dimensional noise, and ill-conditioned covariance matrices—were addressed by implementing targeted 15-channel sensory-motor selection, tight temporal cropping (0.5–3.5 s), Ledoit-Wolf regularized Common Spatial Patterns (CSP), and linear Support Vector Machines (SVM). Across three evaluated subjects, the pipeline achieved holdout test accuracies of 94.44%, 72.22%, and 50.00%. Furthermore, alternative cross-validation paradigms to lower estimator variance are evaluated, intuitive models for CSP and SVM are provided, and recent healthcare privacy debates (e.g., Doctolib's opt-in data practices) are analyzed to formulate a secure, GDPR/EDPB-compliant Federated Learning deployment framework.

1 Introduction and Objectives

Brain-Computer Interfaces (BCIs) establish a direct communication pathway between the central nervous system and external software or hardware devices. A prevalent non-invasive paradigm is *Motor Imagery* (MI), wherein a subject mentally simulates a physical movement without executing muscle activation. This process induces localized power suppression in sensorimotor rhythms (μ: 8–12 Hz and β: 13–30 Hz), known as Event-Related Desynchronization (ERD), over the contralateral motor cortex.

The primary objectives of this project were to:

1. Construct an end-to-end signal processing pipeline using `mne-python` to clean and segment multi-run EEG data from the PhysioNet dataset.
2. Apply spatial channel reduction and regularized feature extraction to mitigate high-dimensional noise and covariance instability.
3. Train and evaluate a linear Support Vector Machine (SVM) classifier across individual subjects.
4. Develop an interactive dark-mode Graphical User Interface (GUI) using `CustomTkinter` and embedded `Matplotlib` dashboards to display real-time signals and probabilities.
5. Analyze system-level engineering hurdles, model evaluation strategies, and privacy-preserving multi-hospital deployment architectures in light of current European healthcare data regulations.

2 Methodology and Pipeline Architecture

2.1 Data Acquisition and Expansion

Data was acquired from the PhysioNet EEGMMIDB database (64 channels recorded at 160 Hz). To resolve the severe sample starvation of single 15-trial runs, 6 execution and imagery runs (Runs 3, 4, 7, 8, 11, and 12) were concatenated per subject. This expanded the total dataset size to $N = 90$ trials per subject (45 Left Hand, 45 Right Hand).

2.2 Preprocessing and Temporal Slicing

To isolate sensorimotor activity and eliminate ocular/muscular artifacts, continuous EEG signals were bandpass filtered between 8.0 Hz and 30.0 Hz. Signals were segmented into 3-second epochs tightly bound between $t = 0.5$ s and $t = 3.5$ s post-cue stimulus. This temporal cropping excludes the variable reaction-latency phase (0.0–0.5 s) to focus exclusively on sustained ERD.

2.3 Sensory-Motor Channel Selection

Rather than using all 64 scalp electrodes—which introduces volume conduction noise from non-motor regions—15 sensory-motor channels covering the fronto-central, central, and centro-parietal strips were isolated:

$$\mathcal{C}_{\text{sub}} = \{FC_5, FC_3, FC_1, FC_z, FC_2, FC_4, FC_6, C_5, C_3, C_z, C_4, C_6, CP_3, CP_z, CP_4\}$$

This spatial constraint forces feature extraction algorithms to focus strictly on C_3 (left motor cortex) and C_4 (right motor cortex).

2.4 Feature Extraction: Regularized CSP

Common Spatial Patterns (CSP) seeks spatial filters W that maximize the variance of the filtered signal for one class while simultaneously minimizing it for the other. Given normalized class covariance matrices Σ_0 and Σ_1:

$$\Sigma_c = \frac{XX^T}{\text{Tr}(XX^T)} \tag{1}$$

To prevent ill-conditioned covariance estimation caused by low sample-to-channel ratios, **Ledoit-Wolf Shrinkage** regularization was applied:

$$\Sigma_{\text{shrunk}} = (1-\gamma)\Sigma_c + \gamma\mu I \tag{2}$$

where I is the identity matrix and $\gamma \in [0, 1]$ is the analytically derived shrinkage intensity. The generalized eigenvalue problem $\Sigma_0 W = \lambda(\Sigma_0 + \Sigma_1)W$ is solved, and the $d = 4$ extreme spatial components are selected to project the matrix down to feature vectors $f \in \mathbb{R}^4$.

2.5 Classification: Linear SVM

The extracted feature vectors feed into a linear Support Vector Machine, which constructs an optimal hyper-plane separating Left and Right motor intent:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{N}\xi_i \quad \text{s.t.} \quad y_i(w^T f_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \tag{3}$$

Table 1: Subject-Specific Motor Imagery Classification Performance

Subject	**Total Trials (N)**	**Train Size (n_{train})**	**Test Size (n_{test})**	**Test Accuracy**
Subject 1	90	72	18	**72.22%**
Subject 2	90	72	18	**94.44%**
Subject 3	90	72	18	**50.00%**

3 Experimental Results and Model Evaluation

Evaluating the pipeline on an 80/20% stratified holdout split ($n_{\text{train}} = 72$, $n_{\text{test}} = 18$) yielded the subject-specific performances detailed in Table 1.

3.1 Discussion of Subject Performance

- **Subject 2 (**94.44%**):** Demonstrates exceptional sensorimotor rhythm modulation. C_3/C_4 desynchronization is pronounced and linearly separable under CSP transformation.
- **Subject 1 (**72.22%**):** Shows robust discriminative ability, performing well above the theoretical statistical chance threshold (50.00% for binary classification).
- **Subject 3 (**50.00%**):** Illustrates the well-documented phenomenon of **BCI Illiteracy** (affecting 15–20% of subjects in BCI literature). Standard static 8–30 Hz bandpass filters fail for these individuals because personal μ/β frequency shifts fall into non-standard sub-bands (e.g., 12–14 Hz).

3.2 Limitations of Holdout Validation and Alternative Evaluation Strategies

While a single 80/20% train-test split provides a fast benchmark, evaluating models on small test sets ($n_{\text{test}} = 18$) introduces high variance in the error estimator $\widehat{R}(f)$, where $\text{Var}(\widehat{R}(f)) \propto \frac{1}{n_{\text{test}}}$. A single misclassified trial alters accuracy by 5.55%.

To achieve more statistically sound validation in future work, two alternative strategies are recommended:

1. **Stratified K-Fold Cross-Validation ($K = 5$ or 10):** Partitions the 90 trials into K equal folds, iteratively training on $K - 1$ folds and testing on the remaining fold. Averaging results across all iterations ensures every trial is evaluated, drastically reducing generalization variance.
2. **Leave-One-Run-Out (LORO) Validation:** Trains the pipeline on 5 recorded runs and tests strictly on the 6th unseen run. This tests temporal stability and resistance to signal drift across recording sessions.

4 System Challenges and Engineering Hurdles

During development, several software engineering and system architecture bottlenecks were resolved, as summarized in Table 2.

5 Privacy, Ethics, and Enterprise Multi-Hospital Deployment

Deploying BCI systems in clinical settings requires navigating stringent European data protection frameworks, specifically GDPR Article 9 regarding biometric special-category data and European Data Protection Board (EDPB) directives.

Table 2: Engineering Bottlenecks and Applied Technical Solutions

Problem	Root Cause	Technical Solution Applied
System Bus Errors (`SIGBUS`)	Native C-extension memory exceptions in compiled SciPy/Pandas binaries on macOS arm64.	Rebuilt environment using native ARM64 architecture Python binaries.
GUI Thread & Terminal Freezes	Matplotlib canvas thread battles with CustomTkinter loop; MNE CLI prompts locking `readline`.	Enforced `matplotlib.use("TkAgg")` and bypassed prompts via `update_path=True`.
High Performance Variance	Overfitting due to 64 non-motor channels introducing ocular/facial noise.	Restricted feature extraction to 15 sensory-motor channels ($FC_{3..4}, C_{3..4}, CP_{3..4}$).
Covariance Matrix Instability	Low trial counts causing ill-conditioned, singular covariance matrices in CSP.	Implemented Ledoit-Wolf Shrinkage regularization (`reg='ledoit_wolf'`).
Visual Cue Latency Artifacts	Idle subject state during visual cue presentation (0–0.5 s) diluted ERD.	Applied tight temporal cropping to $t \in [0.5, 3.5]$ s post-cue stimulus.

5.1 The Healthcare Data Privacy Dilemma: Case Study of Doctolib

A recent case study in healthcare AI governance involved Doctolib's research lab rollout, which faced heavy criticism from privacy advocacy groups (e.g., LDH) and regulators for deploying AI features under an implicit **Opt-Out** scheme for millions of users. GDPR Article 9 mandates that special-category health and biometric data require **explicit, affirmative consent (Opt-In)**.

While explicit Opt-In resolves legal non-compliance, relying *solely* on Opt-In creates technical limitations for medical AI:

- **Data Scarcity and Selection Bias:** Opt-in rates for medical data sharing are historically low (10–15%). Skewed training distributions exacerbate phenomena like BCI illiteracy because models fail to learn diverse neurophysiological profiles.
- **Re-identification Risks:** EEG signals act as unique "brain fingerprints." Standard pseudonymization fails because raw time-series signals can often be re-identified using correlation matching algorithms.

5.2 Architectural Solution: Opt-In Combined with Privacy-Enhancing Technologies (PETs)

To achieve full compliance without sacrificing model performance, enterprise deployment must combine explicit Opt-In consent with a **Federated Learning (FL)** framework using Privacy-Enhancing Technologies:

1. **On-Premises Isolation:** Raw EEG data processed by MNE and CSP remains strictly within the hospital firewall. Raw signals are never transmitted off-site.
2. **Federated Averaging (`FedAvg`):** Participating hospitals train local CSP+SVM instances on local patient data. Only encrypted weight updates (w_{local}) are sent to a central coordinator to compile a global master model (W_{global}).

3. **Differential Privacy (DP):** Calibrated Gaussian noise is injected into local CSP covariance metrics to ensure formal mathematical anonymity (ϵ, δ), preventing gradient inversion attacks.

4. **Homomorphic Encryption (HE):** Model parameter updates are encrypted during transit, allowing the central server to compute global averages over encrypted payloads.

6 Conclusion

This study demonstrated a complete end-to-end motor imagery BCI pipeline with an interactive GUI dashboard. By mitigating high dimensionality and covariance instability through regularized CSP, temporal cropping, and channel selection, high subject classification performance (94.44% top accuracy) was achieved. Finally, it was shown that while explicit Opt-In consent (as highlighted by the Doctolib debate) provides necessary legal protection, technical privacy guarantees require combining Opt-In with Federated Learning and Differential Privacy for secure multi-hospital deployment.